



UNIVERSIDAD NACIONAL DE INGENIERÍA

Facultad de Ciencias

Escuela Profesional de Ciencia de la Computación

CUARTA PRÁCTICA CALIFICADA

Curso: Fundamentos de Programación CC112 - A

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Código: 20232597K

1. Dos Pruebas

1.1

```
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danieldiazm-maker/Semana5/PC4/Mejora_Examen$ make valgrind
mkdir -p build/debug
g++ -std=c++17 -Wall -Wextra -Wpedantic -g -O0 Pregunta01.cpp -o build/debug/Pregunta01
valgrind --leak-check=full --show-leak-kinds=all ./build/debug/Pregunta01
==5775== Memcheck, a memory error detector
==5775== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==5775== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==5775== Command: ./build/debug/Pregunta01
==5775==
Polinomio 1:  $1x^2 + 3$ 
Polinomio 2:  $-4x^1 + 5$ 
Polinomio 3:  $3x^2 + 1$ 
Suma de los tres polinomios:  $4x^2 - 4x^1 + 9$ 
Producto de los tres polinomios:  $-12x^5 + 15x^4 - 40x^3 + 50x^2 - 12x^1 + 15$ 
==5775==
==5775== HEAP SUMMARY:
==5775==    in use at exit: 0 bytes in 0 blocks
==5775==   total heap usage: 9 allocs, 9 frees, 74,888 bytes allocated
==5775==
==5775== All heap blocks were freed -- no leaks are possible
==5775==
==5775== For lists of detected and suppressed errors, rerun with: -s
==5775== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danieldiazm-maker/Semana5/PC4/Mejora_Examen$
```

1.2

```
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danieldiazm-maker/Semana5/PC4/Mejora_Examen$ make valgrind
mkdir -p build/debug
g++ -std=c++17 -Wall -Wextra -Wpedantic -g -O0 Pregunta01.cpp -o build/debug/Pregunta01
valgrind --leak-check=full --show-leak-kinds=all ./build/debug/Pregunta01
==44872== Memcheck, a memory error detector
==44872== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==44872== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==44872== Command: ./build/debug/Pregunta01
==44872==
Polinomio 1:  $-5x^2 + 28x^1 + 37$ 
Polinomio 2:  $25x^3 - 5x^2 - 10x^1$ 
Polinomio 3:  $-38x^3 + 26x^2 - 14x^1 - 2$ 
Suma de los tres polinomios:  $-13x^3 + 16x^2 + 4x^1 + 35$ 
Producto de los tres polinomios:  $4750x^8 - 30800x^7 - 11130x^6 + 29480x^5 - 11170x^4 - 4780x^3 + 6110x^2 + 740x^1$ 
==44872==
==44872== HEAP SUMMARY:
==44872==    in use at exit: 0 bytes in 0 blocks
==44872==   total heap usage: 9 allocs, 9 frees, 74,924 bytes allocated
==44872==
==44872== All heap blocks were freed -- no leaks are possible
==44872==
==44872== For lists of detected and suppressed errors, rerun with: -s
==44872== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

2. Dos Pruebas

2.1

```
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiazm-maker/Semana5/PC4/M
ejora_Examen$ valgrind ./build/debug/Pregunta02
==9750== Memcheck, a memory error detector
==9750== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==9750== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==9750== Command: ./build/debug/Pregunta02
==9750==
Seleccion      PG      PE      PP      GF      GC      DG      Pts      Rend(%)
Pais_3         49      14      37      260     235     25      161     53.7
Pais_2         44      21      35      273     262     11      153     51
Pais_5         46      12      42      258     257     1       150     50
Pais_4         44      15      41      248     249     -1      147     49
Pais_1         38      15      47      259     273     -14     129     43
CAMPEON: Pais_3 | Puntaje: 161 | DG: 25 | Rendimiento: 53.7%
==9750==
==9750== HEAP SUMMARY:
==9750==   in use at exit: 0 bytes in 0 blocks
==9750== total heap usage: 3 allocs, 3 frees, 74,952 bytes allocated
==9750==
==9750== All heap blocks were freed -- no leaks are possible
==9750==
==9750== For lists of detected and suppressed errors, rerun with: -s
==9750== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiazm-maker/Semana5/PC4/M
ejora_Examen$
```

2.2

```
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiazm-maker/Semana5/PC4/M
ejora_Examen$ valgrind ./build/debug/Pregunta02
==45163== Memcheck, a memory error detector
==45163== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==45163== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==45163== Command: ./build/debug/Pregunta02
==45163==
Seleccion      PG      PE      PP      GF      GC      DG      Pts      Rend(%)
Pais_3         42      19      39      236     245     -9      145     48.3
Pais_5         42      14      44      241     237     4       140     46.7
Pais_4         40      20      40      261     271    -10      140     46.7
Pais_1         39      19      42      245     255    -10      136     45.3
Pais_2         38      15      47      245     253     -8      129     43
CAMPEON: Pais_3 | Puntaje: 145 | DG: -9 | Rendimiento: 48.3%
==45163==
==45163== HEAP SUMMARY:
==45163==   in use at exit: 0 bytes in 0 blocks
==45163== total heap usage: 3 allocs, 3 frees, 74,952 bytes allocated
==45163==
==45163== All heap blocks were freed -- no leaks are possible
==45163==
==45163== For lists of detected and suppressed errors, rerun with: -s
==45163== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

3. Dos Pruebas

3.1

```
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-m-maker/Semana5/PC4/M
ejora_Examen$ valgrind ./build/debug/Pregunta03
==20285== Memcheck, a memory error detector
==20285== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==20285== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==20285== Command: ./build/debug/Pregunta03
==20285==
INVENTARIO
Codigo: 100 | Nombre: Teclado | Precio: 10.5 | Stock: 5
Codigo: 101 | Nombre: Mouse | Precio: 21 | Stock: 10
Codigo: 102 | Nombre: Monitor | Precio: 31.5 | Stock: 15
Codigo: 103 | Nombre: Laptop | Precio: 42 | Stock: 20
Codigo: 104 | Nombre: Impresora | Precio: 52.5 | Stock: 25

Buscando producto con codigo 102...
Producto encontrado: Monitor | Precio: 31.5
==20285==
==20285== HEAP SUMMARY:
==20285==   in use at exit: 0 bytes in 0 blocks
==20285==   total heap usage: 8 allocs, 8 frees, 74,951 bytes allocated
==20285==
==20285== All heap blocks were freed -- no leaks are possible
==20285==
==20285== For lists of detected and suppressed errors, rerun with: -s
==20285== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-m-maker/Semana5/PC4/M
ejora_Examen$
```

3.2

```
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-m-maker/Semana5/PC4/M
ejora_Examen$ valgrind ./build/debug/Pregunta03
==46592== Memcheck, a memory error detector
==46592== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==46592== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==46592== Command: ./build/debug/Pregunta03
==46592==
INVENTARIO
Codigo: 100 | Nombre: Procesador | Precio: 975 | Stock: 12
Codigo: 101 | Nombre: Tarjeta Grafica | Precio: 1650 | Stock: 18
Codigo: 102 | Nombre: Monitor | Precio: 275 | Stock: 15
Codigo: 103 | Nombre: Laptop | Precio: 2050 | Stock: 20
Codigo: 104 | Nombre: Impresora | Precio: 675 | Stock: 25

Buscando producto con codigo 104...
Producto encontrado: Impresora | Precio: 675
==46592==
==46592== HEAP SUMMARY:
==46592==   in use at exit: 0 bytes in 0 blocks
==46592==   total heap usage: 8 allocs, 8 frees, 74,964 bytes allocated
==46592==
==46592== All heap blocks were freed -- no leaks are possible
==46592==
==46592== For lists of detected and suppressed errors, rerun with: -s
==46592== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

4. Dos pruebas

4.1

```
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danieldiazm-maker/Semana5/PC4/Mejora_Examen$ valgrind ./build/debug/Pregunta04
==43697== Memcheck, a memory error detector
==43697== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==43697== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==43697== Command: ./build/debug/Pregunta04
==43697==
Pedido N°: 101
Cliente: Carlos Perez

Items:
- Laptop | Cant: 1 | Precio: 100
- Mouse | Cant: 2 | Precio: 200
- Teclado | Cant: 3 | Precio: 300

Total: 1400

Item mas caro: Teclado | Precio: 300
==43697==
==43697== HEAP SUMMARY:
==43697==    in use at exit: 0 bytes in 0 blocks
==43697== total heap usage: 8 allocs, 8 frees, 74,890 bytes allocated
==43697==
==43697== All heap blocks were freed -- no leaks are possible
==43697==
==43697== For lists of detected and suppressed errors, rerun with: -s
==43697== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danieldiazm-maker/Semana5/PC4/Mejora_Examen$
```

4.2

```
ejora_Examen$ valgrind ./build/debug/Pregunta04
==47132== Memcheck, a memory error detector
==47132== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==47132== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==47132== Command: ./build/debug/Pregunta04
==47132==
Pedido N°: 192
Cliente: Daniel Diaz

Items:
- Procesador | Cant: 3 | Precio: 975
- Monitoe | Cant: 3 | Precio: 275
- Tarjeta Grafica | Cant: 3 | Precio: 1600

Total: 8550

Item mas caro: Tarjeta Grafica | Precio: 1600
==47132==
==47132== HEAP SUMMARY:
==47132==    in use at exit: 0 bytes in 0 blocks
==47132== total heap usage: 8 allocs, 8 frees, 74,903 bytes allocated
==47132==
==47132== All heap blocks were freed -- no leaks are possible
==47132==
==47132== For lists of detected and suppressed errors, rerun with: -s
==47132== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danieldiazm-maker/Semana5/PC4/Mejora_Examen$
```